

ECO2010H1F
Mathematics and Statistics for PhD Students
University of Toronto

Department of Economics
Instructor: Prof. M. Burda

Term Test 1

Date: August 22, 2025

Time: 10:10 am - 12:00 pm

Formula Sheet

Definition 1 An integer x is called **even** (respectively **odd**) if there is another integer k for which $x = 2k$ (respectively $2k + 1$).

Remark 1 An implication and its contrapositive are equivalent statements:

$$\{\mathcal{A} \implies \mathcal{B}\} \iff \{\neg \mathcal{B} \implies \neg \mathcal{A}\}$$

Definition 2 Let X be a non-empty set. A **metric** on X is a function $d : X \times X \rightarrow \mathbb{R}_+$ with the following properties:

- (i) $d(x, y) = 0$ iff $x = y$;
- (ii) $d(x, y) = d(y, x)$ for all $x, y \in X$ (Symmetry);
- (iii) $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$ (Triangle Inequality).

Definition 3 If (X, d) is a metric space and $x \in X^{\mathbb{N}}$, then

- (1) x is a bounded sequence iff $x(\mathbb{N})$ is bounded,
- (2) x is a Cauchy sequence iff for all $r \in \mathbb{R}_+$ there exists $N \in \mathbb{N}$ such that, for all $n, m > N$,

$$d(x_m, x_n) < r$$

- (3) x is a convergent sequence iff there exists a limit point $\bar{x} \in X$ such that, for all $r \in \mathbb{R}_+$, $x^{-1}(X \setminus B_r(\bar{x}))$ is finite.

Definition 4 A metric space X is said to be **complete** if every Cauchy sequence in X converges to a point in X .

Definition 5 Suppose that V is a finite-dimensional real inner product space. A basis $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ of V is said to be an **orthogonal** basis of V if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for every $i, j = 1, \dots, r$ satisfying $i \neq j$. It is said to be an **orthonormal** basis if it satisfies the extra condition that $\|\mathbf{v}_i\| = 1$ for every $i = 1, \dots, r$.

Definition 6 Suppose that $\mathbf{v}_1, \dots, \mathbf{v}_r$ are vectors in a vector space V over $\mathbb{R}$.

(LD) We say that $\mathbf{v}_1, \dots, \mathbf{v}_r$ are **linearly dependent** if there exist $c_1, \dots, c_r \in \mathbb{R}$, not all zero, such that $c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r = 0$.

(LI) We say that $\mathbf{v}_1, \dots, \mathbf{v}_r$ are **linearly independent** if they are not linearly dependent; in other words, if the only solution of $c_1 \mathbf{v}_1 + \dots + c_r \mathbf{v}_r = 0$ in $c_1, \dots, c_r \in \mathbb{R}$ is given by $c_1 = \dots = c_r = 0$.

Definition 7 Let X and Y be metric spaces. A correspondence $\Gamma : X \rightrightarrows Y$ is **upper hemicontinuous at** x_0 iff for all open sets V such that $\Gamma(x_0) \subseteq V$, there exists an open set $U \ni x_0$ s.t. $\forall x \in U, \Gamma(x) \subseteq V$.

Γ is **upper hemicontinuous** iff it is upper hemicontinuous at all $x \in X$.

Definition 8 Let X and Y be metric spaces. A correspondence $\Gamma : X \rightrightarrows Y$ is **lower hemicontinuous at** x_0 iff for all open sets V such that $V \cap \Gamma(x_0) \neq \emptyset$, there exists an open set $U \ni x_0$ s.t. $\forall x \in U, V \cap \Gamma(x) \neq \emptyset$.

Γ is **lower hemicontinuous** iff it is lower hemicontinuous at all $x \in X$.

The Bellman equation is given by

$$v_t(s_t) = \max_{a_t \in A_t} \{f_t(a_t, s_t) + \beta v_{t+1}(s_{t+1}) : s_{t+1} = g_t(a_t, s_t)\}$$

The Hamiltonian is given by

$$H_t(a_t, s_t, \lambda_{t+1}) = f_t(a_t, s_t) + \beta \lambda_{t+1} g_t(a_t, s_t)$$

Optimality requires

$$\begin{aligned} \frac{\partial}{\partial a_t} H_t(a_t, s_t, \lambda_{t+1}) &= 0, \quad t = 0, 1, \dots, T-1 \\ \frac{\partial}{\partial s_t} H_t(a_t, s_t, \lambda_{t+1}) &= \lambda_t, \quad t = 1, 2, \dots, T-1 \\ \lambda_T &= v'(s_T) \end{aligned}$$

Term Test Questions:

Question 1 (15 points)

Prove the following Theorem by contrapositive.

Theorem 1 *If n^2 is even for some integer n , then n is even.*

Question 2 (20 points)

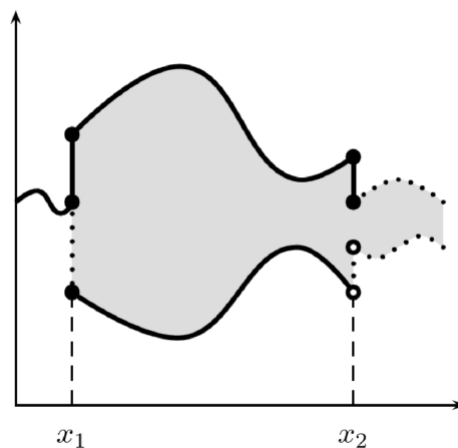
Let (x^m) and (y^m) be two Cauchy sequences in a metric space (X, d) . Prove that the sequence $\{d(x_m, y_m)\}_{m \in \mathbb{N}} \subseteq \mathbb{R}$ converges, noting that $\mathbb{R}$ is a complete set.

Question 3 (20 points)

Prove the following Proposition:

Proposition 1 *Suppose that the non-zero vectors $v_1, \dots, v_r$ in a finite-dimensional real inner product space are pairwise orthogonal. Then they are linearly independent.*

Question 4 (15 points) Is the following correspondence upper hemicontinuous, lower hemicontinuous, neither, or both? Explain your reasoning.



Question 5 (20 points)

Consider a dynamic optimization problem with $T = 2$, no uncertainty, where a household maximizes the lifetime utility

$$U(C_0, C_1) = C_0^\alpha + \beta C_1^\alpha.$$

with parameter $\alpha \in (0, 1)$ and a discount factor $\beta \in (0, 1)$. Period- t income is deterministic and denoted by w_t . Assets evolve as

$$A_0 = 0 \quad (1)$$

$$A_1 + C_0 = w_0 + (1 + r)A_0, \quad (2)$$

$$A_2 + C_1 = w_1 + (1 + r)A_1 \quad (3)$$

and there is no bequest motive (A_2 is free but optimally set to zero).

- (a) Solve the last-period problem and write the value function at $t = 1$.
- (b) Use backward induction to solve for C_0^* and C_1^* in closed form.
- (c) Report A_1^* and verify that the allocation exhausts the intertemporal budget.

Question 6 (10 points)

Solve the following ODEs:

- (a) First order separable in t ,

$$\frac{dx}{dt} = 2t \quad \text{with} \quad x(0) = 1$$

- (b) First order separable

$$\frac{dx}{dt} = kx \quad \text{with} \quad x(0) = 5, \quad k > 0$$